Track 2 report — from a confirmed genotype to graded, approved medicines, with its own negative control

Standing: draft, written 2026-09-19, revised 2026-09-20 from executed runs. Iteration 2 of Track 2 supersedes iteration 1 (never uploaded): the mechanism card is at **version 5** with all nine measured facts filled — the last one, the aneuploid fraction in the sequenced sample, by iteration 2's step 8, now on **two instruments that share nothing past the raw reads** — and the report tells the whole path, steps 1 to 8, plus a **transfer run** of the same pipeline on a second, different case. Every number is read from a table with a provenance manifest; every claim was registered before its test; every cited paper was resolved against PubMed. **Not uploaded.** Track 2 accepts one submission (due about 2026-10-20).

Written for a panel that includes clinicians and patient advocates as well as scientists. Each technical term appears once with its meaning in brackets; the full dictionary is served at `/learn` on the dashboard, and the same tables render on its `/track2` page.

> The challenge's own words, kept in view throughout: *submissions are hypotheses for follow-up, not evidence that a medicine works.* Nothing below is a recommendation to treat.

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1. How this report answers the panel's four questions

> **Rubric:** all four — this is the map; every row names the section that carries the work and the softest spot we know in it.

criterion	weight	where	the evidence, in one line	the weakest point we know
Scientific rigor — is the mechanism sound, and is the drug argument built on it?	35 %	§2–§7	a genotype confirmed by the challenge's own answer key and then **rediscovered blind**; nine measured facts on a versioned mechanism card, the last on two independent instruments; every approved drug graded on a ladder written **before** the search, with a decoy gene through the same pipeline as a negative control	the two rungs that carry the argument (a result in an animal, a result in a dish) are **13 abstracts read by one person on one day**; the deepest mechanistic lead is a single mouse paper, inherited by two vitamin forms; **no** drug has a molecule-level animal result on the checkpoint itself
Potential impact — what could this change if it validated?	25 %	§8	a do-not table that turns the mechanism into one concrete question for the treating team; a mechanism-level (not gene-level) conclusion that reaches the other MVA genes; a **sequencing-only** bound on mosaic aneuploidy with a stated detection limit, where the classical test needs cells that grow	nothing here has been tested in this child or in any patient; the do-not table weighs a mechanism hypothesis against regimens that have survival data behind them; the aneuploidy bound is a statement about a sample whose **tissue the delivery never named**
Innovation — is the angle genuinely new?	25 %	§9	one structured card drives every query; a pre-registered evidence ladder; a decoy gene *and* a whole transfer case as controls; a do-not table as a first-class output; an append-only claim register where each claim carries its refutation condition and corrections stay visible	every part is standard on its own — public APIs, a rubric, a register. What we claim is the **assembly**, and only "to our knowledge": we did not run a literature survey. Knowledge-graph repurposing rankers exist (PrimeKG, TxGNN — §9) and we did not run them
Scalability — does this reach other people and other diseases?	15 %	§10	the pipeline was **executed unchanged** on a transfer case with a different MVA gene, a different disease, a different tumour and its own phenotype terms read out of a public annotation file: a second complete output, at **0 lines of code changed** and 0 hand-typed queries	our own pre-registered claim about the transfer came back **refuted** (H066 · 0.10): four limbs held, including every limb that tests portability, but the spindle-poison caution **did not fire** on the transfer case, because the public pathway database had no annotation for that gene. So the **automated safety half degrades on a new gene with thin pathway coverage** — a new case needs that coverage checked before its do-not table is trusted (§10.5). Also: one transfer case, sharing the same pathway; only the **automated** rungs compare, because a transfer card has no curated literature, and the curated rungs that carry the drug argument stay a human bottleneck

2. The answer in one page

> **Rubric:** scientific rigor (35 %) — the claim, the confirmed genotype behind it, and what is not known, on one page.

The child. A boy with mosaic variegated aneuploidy (MVA: some of his cells carry a wrong number of chromosomes, and different cells are wrong in different ways), short stature, failure to thrive, calcium deposits in the kidneys, and a rhabdomyosarcoma (a childhood cancer of muscle tissue). Eight clinical terms were delivered with the case; no karyotype was.

The cause, found twice. Track 1 found two different broken copies of one gene, `BUB1B` — one from each parent (a *compound heterozygote*) — and the challenge's answer key confirmed the pair with a full match at rank 1. Iteration 2 then rebuilt the answer **blind**, from the raw reads and the eight clinical terms, by routes sharing no code with the first: the same pair came out first. The gene makes BubR1, the **brake** on cell division that holds a dividing cell until every chromosome is attached. One copy makes nothing (a *stop* change); the other makes a weakened protein (a *missense* change in the protein's last stretch). Less brake, not no brake — the only pattern compatible with life, and the pattern every published case shows.

Nine measured facts sit on the mechanism card (now at version 5) so that no drug argument rests on the allele labels alone. The one that matters most for this report: **no chromosome in the sequenced sample carries a mosaic population above 2–3 % of cells** (step 8, at 80 % power), where the statistic used earlier would have found nothing at any fraction. That bound is now carried by **two instruments that share nothing past the raw reads** — allele balance, and a k-mer dosage route that reads the sex chromosomes the first is blind to, recovering one copy of X (0.4873) and one of Y (0.5083) against 0.5 expected as its positive control, and setting an honest floor of **1.6 % of cells** on what it could have detected. It is a statement about the sequenced sample — its tissue is not named by the delivery, and a karyotype's fraction is measured in cultured cells — not about the diagnosis.

Four ways a drug could act, graded on a five-rung ladder written before the search, then run again with the gene swapped for two decoys as a control:

Direction	The idea	What we found	Standing after step 8
D1 reinforce the brake	strengthen the remaining checkpoint	**no approved drug does this**; every checkpoint drug on the market or in trials *weakens* it	the do-not table
D3 raise the amount of working BubR1	the weakened copy still works — make more of it last longer	**niacin / nicotinamide** (vitamin B3 forms): boosting the NAD^+ pathway raised BubR1 in mice bred with too little of it (rung 3, animal model, by proxy); valproic acid and vorinostat act on BubR1's acetylation switch (rung 2, class level)	**the deepest mechanistic lead** — acts on the constitutional genotype whatever the burden; thinnest automated support
D4 the child's actual tumour	approved, non-chemotherapy drugs with a mechanism in rhabdomyosarcoma	**statins**: rhabdomyosarcoma cells depend on making their own cholesterol; lovastatin itself slowed them (rung 2, molecule level); simvastatin has been in a completed paediatric sarcoma trial (NCT02390843)	**the best-supported lead** — a molecule-level result and a paediatric trial agree
D2 exploit the stress of aneuploid cells	aneuploid cells are already stressed; some approved drugs tip them over	**chloroquine** preferentially killed aneuploid cells in the reference screen (rung 2, molecule level); metformin, hydroxychloroquine, bortezomib inherit class results	**literature only** — the sequenced sample shows no aneuploid population ≥ 2–3 % for such a drug to act on; a tumour-context angle, not a treatment of the child's state

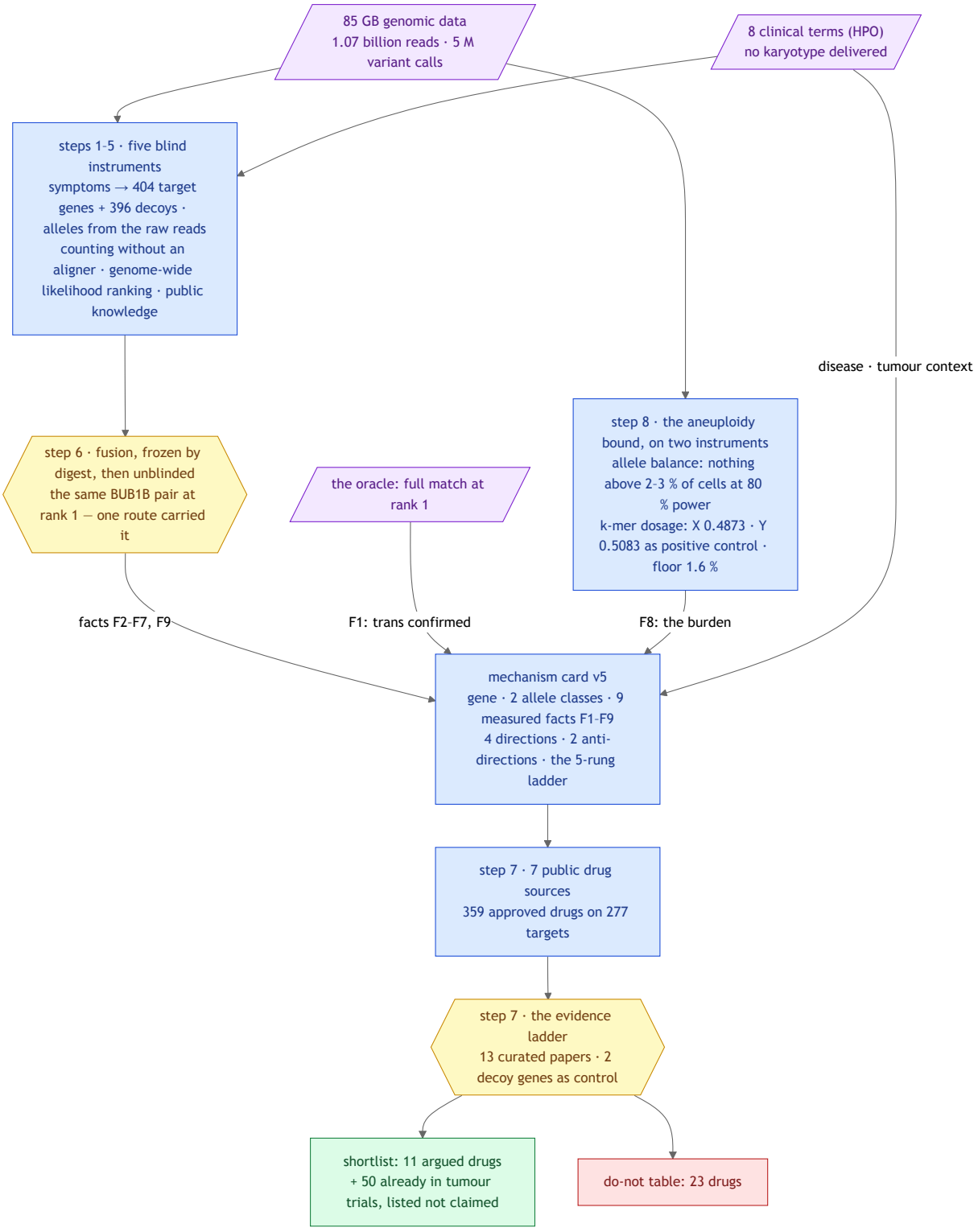
What not to do — the half a clinician may care about most. Thirteen approved drugs are *spindle poisons*: they jam cell division and rely on the checkpoint to hold the cell. In a child whose checkpoint is the broken part, that reliance is a hypothesis worth weighing. Vincristine and vinorelbine are standard rhabdomyosarcoma chemotherapy. **This report flags the question for the treating team; it does not advise withholding standard care.** No approved drug weakens the checkpoint further — that class exists only in trials, and it is the one to avoid.

What we do not know. Whether any drug on this page helps (rung 3 is one mouse paper). Whether the weakened BubR1 copy is unstable or simply inactive. What tissue was sequenced. Anything about events smaller than a chromosome arm, or about a copy-neutral swap that moves no sequence for the dosage route to see.

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3. Two datasets, one line of evidence — the whole path

> **Rubric:** scientific rigor (35 %) — both input datasets, and which stage each one acts on.



> **Legend** — **purple = raw inputs** (the child's DNA, the child's clinical terms, the oracle's reply); **blue = machinery** (every box is a tool with a manifest and a methodology entry); **yellow = decision points** (the fusion that was frozen before it was compared; the ladder whose rungs were written before the search); **green = the goal**; **red = what to avoid**. The labelled arrows are the measured facts each step contributed to the card.

Both datasets act at every stage: the genomic data fixed the gene, the alleles, their dosage and the absence of aneuploidy in the sample; the clinical terms fixed the disease, the target set of step 1, the tumour context and — through the tumour — direction D4 and the paediatric questions. A result that uses only one says so.

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4. How the genotype was established — steps 1 to 6 and 8, in the order they ran

> **Rubric:** scientific rigor (35 %) — how the genotype the drug argument rests on was established and checked.

Iteration 2 was designed as a **blind mirror** of the first answer: could the pair be found again from the raw material, by instruments sharing no code with iteration 1, with no iteration-1 table read until the last entry? It could — and the way it could is the finding.

Step	Question	What it measured	What it gave the drug argument
1 the blind target set	which genes do the eight symptoms alone nominate?	404 genes tiered by symptom evidence, 396 size-matched **decoy genes** beside them; held-out recall@300 **0.335** — a third of known disease genes would be missed. 73.6 % of protein-coding genes have no symptom curation at all	the decoy set that later controls the drug pipeline; the lesson that symptom-only ranking is an *enrichment*, never coverage
2 alleles from the raw reads	what do the reads carry in those genes, without the vendor's caller?	14.7 M reads recovered from 1.07 billion by a k-mer bait; two independent callers agreed with the vendor at 99.6 % / 99.3 %; **1,199 alleles the vendor never reported**; 164 structural variants, **none in the gene**; 6,823 phased blocks, none reaching the gene	F4: no hidden deletion; phase unmeasured in the reads — *trans* is the oracle's statement
3 counting, not aligning	are the alleles at one copy each, and every exon at two?	164,044 alleles typed from k-mer counts with no aligner and no caller; **both lead alleles at one copy** (posterior 1.0); every exon region at **two copies** (median 2.00; the X at 0.95 in a male, found with nothing told to the method)	F2, F3: a *null + weakened* genotype — some BubR1 exists to stabilise, so D3 is live
4 genome-wide likelihood ranking	with no hard filter, which pair has the highest posterior?	**51,920** biallelic hypotheses over 11,465 genes; `BUB1B` first at a ratio of 341,019; the missense's pathogenicity lane 4.33; targets separate from decoys at AUC 0.706; population phasing **undetermined**	F6, F7: "damaging" quantified; population data cannot corroborate *trans*
5 public knowledge	does the literature nominate the pair?	815 genes asked of LitVar, PubTator, UniProt/InterPro, AlphaFold and PubMed; the route separates from decoys (AUC 0.634) and is *not* citation volume; yet its top two genes are decoys and `BUB1B` sits at rank **293** of 815; the missense lies **outside every curated domain**, in a region the structure model is confident about (pLDDT 91)	F5: whether the weakened copy is unstable or inactive is **open**; D3 is argued as *more BubR1*, not *refold it*. The literature is searched per drug, never trusted as a ranker
6 fusion, then unblinding	fused, does the mirror rediscover the pair?	the ranking frozen (sha256 recorded) *before* any iteration-1 table was read; **the same pair at rank 1** under both weightings — but silencing the likelihood route drops it to rank 62; the symptom route had it at rank 9; the literature route never listed it. A second oracle upload: **0.000 bits** of information — not made	F9: a rediscovered, confirmed answer; and the honesty rule the drug section inherits — *one route found it, two corroborated at a distance*
8 the aneuploidy bound, twice	is mosaic aneuploidy detectable in the sample, and at what limit?	**instrument 1, allele balance:** 2,778,844 heterozygous calls reduced to 127,913 arm rows; a fold-over statistic reaches 80 % power at **2 % of cells** (chr1–6) and **3 %** (chr7–22), where the statistic iteration 1 used had no power at any fraction; **no autosome above its own limit**; the divisor holds under three injected abnormal chromosomes. **Instrument 2, k-mer dosage** (shares nothing with the first past the raw reads): reference-unique 31-mers counted over 32 of 272 flowcell shards recover **one copy of X (0.4873) and one of Y (0.5083)** against 0.5 expected — the positive control allele balance cannot give in a	F8: **measured on two instruments**, with X and Y covered and a positive control behind them; D2 becomes literature-only

Step	Question	What it measured	What it gave the drug argument
		male — and put no autosome beyond its composition-corrected null (chromosome-GC regressor, Pearson 0.904, 81.7 % of the spread explained; residual spread 0.802 %); the one chromosome that looked suspicious (chr21, 2.33 σ, Bonferroni p 0.43) dissolved under tail trimming, a probe half-split and a depth check that read the excess as mismapping pileups. Honest floor **1.6 % of cells**, twice the residual spread — not the ±1 % a bootstrap interval suggested	

Three things this path teaches that shape the drug section: the decoys were the most useful thing built (they later exposed a non-specific drug direction); a "negative" result is only one when it carries its detection limit (step 8 corrected step 0's flat karyogram on exactly that point); and a rank-1 answer with one route carrying it is reported as such, not as a consensus.

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5. The mechanism (methods form Q14: loss- versus gain-of-function, pathway, consequence)

> **Rubric:** scientific rigor (35 %) — the variant mechanism characterization the panel asks for: loss- versus gain-of-function, pathway, downstream consequence.

5.1 Loss of function, of a specific shape

copy	what changed	what it does to the protein	evidence
one	a *stop* change in exon 17 of 23	the protein ends early, before its last working part — **no functional protein**	ClinVar VCV000533901, pathogenic / likely pathogenic for MVA1
the other	a *missense* change in exon 23 of 23 — one amino acid swapped	a **weakened** protein: never seen in 140,000 healthy genomes, predicted damaging by two independent predictors; outside every catalogued domain, in a region the structure model is confident about	iteration 1 entries 070–074; iteration 2 entries 099, 106

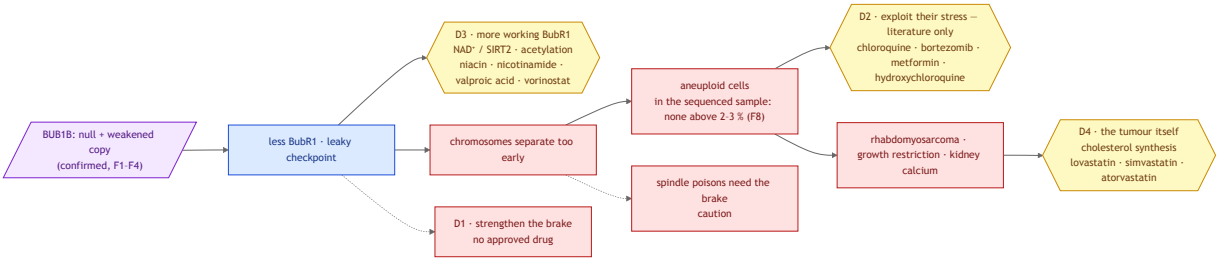
No gain-of-function or dominant-negative activity is reported for `BUB1B`; complete loss is lethal before birth in mouse and human (Hanks 2004, PMID 15475955; Suijkerbuijk 2010, PMID 20516114), which is why "switch BubR1 off" is never a therapeutic direction.

5.2 The nine measured facts on the card (version 5)

fact	measured	what it changes for a drug
F1 the two copies are on different chromosomes (*trans*)	**confirmed** by the answer key (full match, rank 1)	the argument stands on a confirmed genotype
F2 each copy present once in the raw reads	yes — both alleles at one-copy dosage, no aligner	a *null + weakened* genotype: some BubR1 exists (D3 live)
F3 no deleted or duplicated stretch in the gene	every exon at two copies	the two alleles are the whole story
F4 structural variants and read-backed phase	none in the gene; phase unmeasured in the reads	*trans* is the oracle's statement, and the report says so
F5 where the missense sits	outside every catalogued domain, pLDDT 91	unstable or inactive: **open**; D3 is a dosage argument
F6 how damaging the missense is	AlphaMissense likely-pathogenic (lane 4.33); the pair leads 51,920 hypotheses	"damaging" quantified
F7 population phasing	undetermined — 5 of 50 leading alleles reach gnomAD	population data cannot corroborate F1
F8 aneuploid fraction in the sequenced sample	**measured by two instruments that share nothing past the raw reads.** Allele balance: **no autosome above its limit — 2 % (chr1–6), 3 % (chr7–22) at 80 % power**; X and Y unreadable by allele balance in a male. k-mer dosage: X 0.4873 and Y 0.5083 against 0.5 expected as a positive control, no autosome beyond its composition-corrected null, **honest floor 1.6 % of cells**. Still outside both routes: sub-arm events, and a copy-neutral swap that moves no sequence; the tissue is unnamed	**D2 cannot be argued from a burden in this child** — now on two routes, not one; D3, D4 and the do-not table are untouched
F9 the blind mirror	found the same pair first	a rediscovered, confirmed answer

5.3 The pathway, and where a drug could stand

BubR1 is a core part of the **spindle assembly checkpoint** (SAC — the wait-until-attached brake) and of the complex that switches off the machinery pulling chromosomes apart. Less BubR1 means a **leaky brake**: division can proceed with chromosomes still unattached.



> **Legend** — **purple** the genotype; **blue** the molecular consequence; **yellow** the three places an approved drug could stand, each a direction below; **red** what goes wrong, and the two dotted boxes are the direction with no drug and the class to be cautious with.

Downstream, in order: premature separation of chromosomes → the aneuploidy that names the disease (not detectable above 2–3 % in the sample sequenced here) → in surviving aneuploid cells a chronic stress from carrying the wrong number of copies of many proteins → a predisposition to exactly the cancers this child has → growth restriction and nephrocalcinosis, the delivered clinical terms.

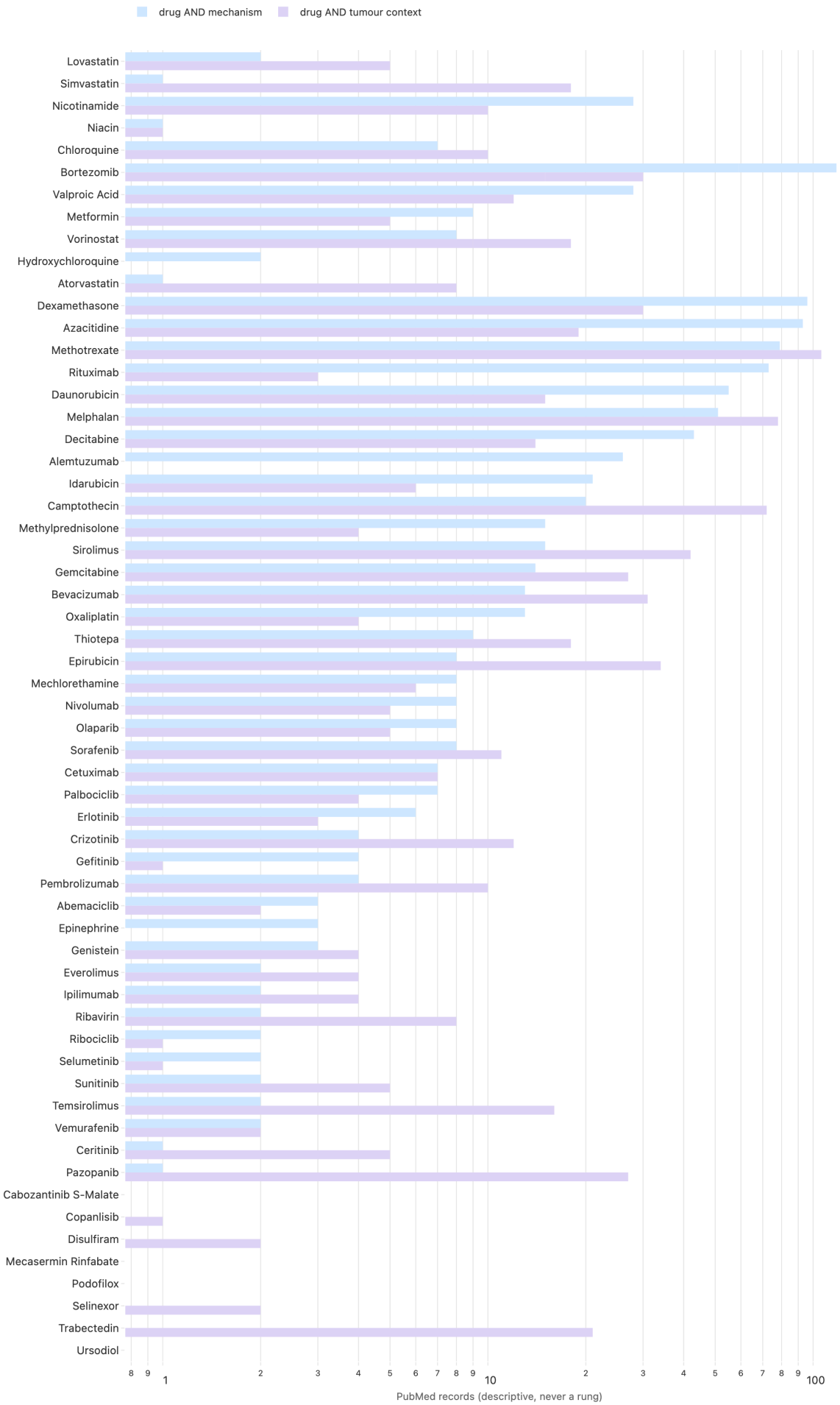
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6. How the candidates were found, graded and controlled

> **Rubric:** scientific rigor (35 %) — how candidates were searched, graded and controlled, with the rules fixed before the search.

6.1 One card, seven public sources, no hand-typed query

The mechanism card drives every query. Reactome widens the card's 32 seed genes to 277 targets; Open Targets, ChEMBL and DGIdb return every drug known to act on them (3,859 rows, **359 market-approved drugs**); openFDA confirms a US label and whether it has a *Pediatric Use* section; ClinicalTrials.gov answers "in this tumour?" and "in children, anywhere?"; PubMed answers "how much has been written about this drug with this mechanism?" — per drug, with the vocabulary read off the card. Every table carries a manifest with the retrieval time and a digest of every reply; the same commands run another case by swapping the card — which is no longer a claim, it was executed (§10).


drug AND mechanism
drug AND tumour context
Lovastatin
Simvastatin
Nicotinamide
Niacin
Chloroquine
Bortezomib
Valproic Acid
Metformin
Vorinostat
Hydroxychloroquine
Atorvastatin
Dexamethasone
Azacitidine
Methotrexate
Rituximab
Daunorubicin
Melphalan
Decitabine
Alemtuzumab
Idarubicin
Camptothecin
Methylprednisolone
Sirolimus
Gemcitabine
Bevacizumab
Oxaliplatin
Thiotepa
Epirubicin
Mechlorethamine
Nivolumab
Olaparib
Sorafenib
Cetuximab
Palbociclib
Erlotinib
Crizotinib
Gefitinib
Pembrolizumab
Abemaciclib
Epinephrine
Genistein
Everolimus
Ipilimumab
Ribavirin
Ribociclib
Selumetinib
Sunitinib
Temsirolimus
Vemurafenib
Ceritinib
Pazopanib
Cabozantinib S-Malate
Copanlisib
Disulfiram
Mecasermin Rinfabate
Podofilox
Selinexor
Trabectedin
Ursodiol
8 9 1 2 3 4 5 6 7 8 9 10 2 3 4 5 6 7 8 9 100
PubMed records (descriptive, never a rung)

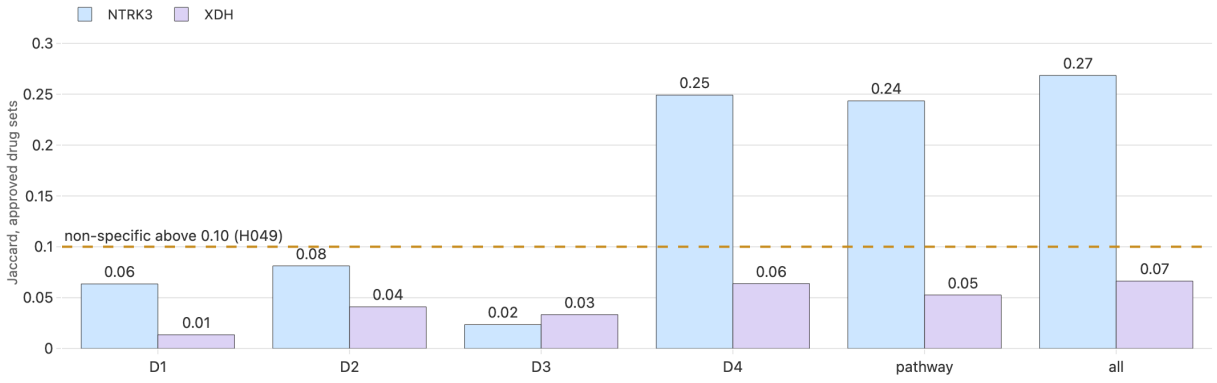
> **Legend** — one row per shortlisted drug (all 61), two bars each, on a **logarithmic** record axis: **blue = the drug searched with this case's mechanism** (the gene, the protein, the pathway or the disease, all taken off the card); **purple = the drug searched with the tumour context**. Missing bar = zero records found. These counts are **descriptive only and never move a rung** — the figure is here to show what the literature volume looks like per drug, and to make one point visible: the best-supported leads — the statins and the vitamin-B3 forms — sit among the *smallest* bars, while the largest belong to drugs this report lists and does not claim (§7.2).

6.2 The ladder, written before the search

rung	meaning	how it is tested
5	approved for a related indication, with paediatric use	approved **and** a registered trial in the child's tumour **and** a paediatric pointer (label section 8.4, or a study admitting children) — **applied to the tumour direction only**
4	clinical trial in a related indication	a registered trial in the child's tumour — tumour direction only
3	animal model of the mechanism	a paper read by a person, with its PubMed id
2	cell model of the mechanism	a paper read by a person, with its PubMed id
1	mechanism only	acts on a target in the direction's set — *named, never submitted*

Automated and curated tests print side by side. Inherited evidence is marked `proxy` with the compound actually tested. **One rule was narrowed after the first run and the change is recorded:** a rhabdomyosarcoma trial was lifting drugs onto rung 5 on the *mechanism* directions (85 drugs, tetracycline among them); a trial in the tumour says nothing about a checkpoint, so trial-based rungs now grade the tumour direction alone (entry 113). The same pre-registered test was then run **twice**, once per card revision: adding the measured aneuploid fraction to the card (v4, entry 120, H061) moved **no rung at all**, and adding step 8's second instrument to it (v5, entry 133, H065) moved **no rung, no shortlist row and no do-not row** — the three table digests came back unchanged. The ladder is independent of F8 by construction; the first run of the test caught a file-reading fault along the way that would otherwise have gone unnoticed, which is why it is re-run for every card revision rather than reasoned about.

6.3 The control: the same child, a different gene



> **Legend** — one bar per decoy gene; the dashed yellow line is the pre-registered 0.10 threshold. **D4 is above it** against NTRK3 (Jaccard 0.25): the cancer pharmacopoeia comes out for any cancer-adjacent gene, and 64 of the decoy's drugs had a rhabdomyosarcoma trial with the gene never consulted. **D1–D3 are below it** against

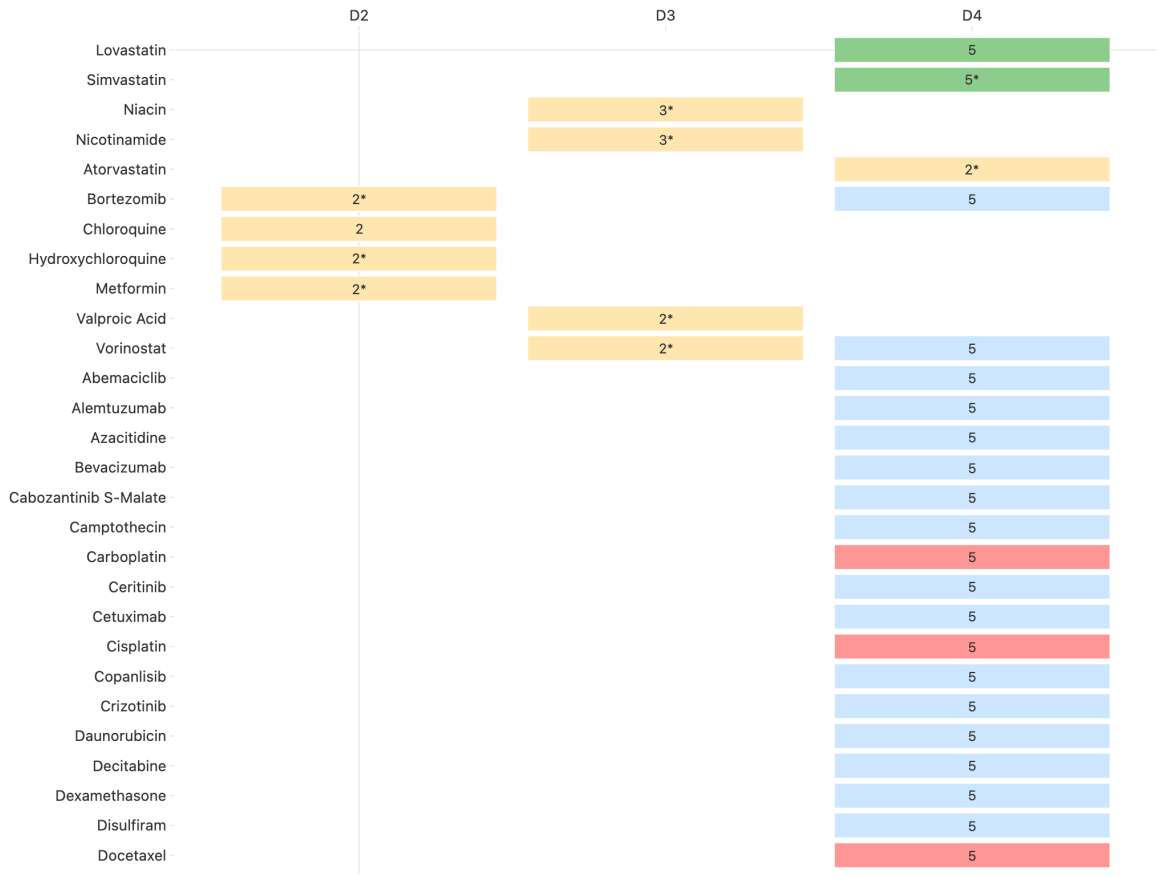
> both decoys. Consequence: D4 is argued only from curated mechanism results, never from "this drug has a rhabdomyosarcoma trial".

The decoys were chosen before the run: NTRK3 by a seeded draw from step 1's 396 matched decoys (a receptor kinase with approved inhibitors — a harder control than a random gene), and XDH, a metabolic enzyme with nothing to do with cell division.

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7. The candidates

> **Rubric:** scientific rigor (35 %) — the proposed repurposing candidates and the evidence each one carries.



> **Legend** — the number is the rung; **green** a curated result and rung 4–5; **yellow** a curated result at rung 2–3; **blue** the automated tests alone (already being tried in the tumour — not a finding of this work); **red** set aside; `*` inherited (proxy) evidence.

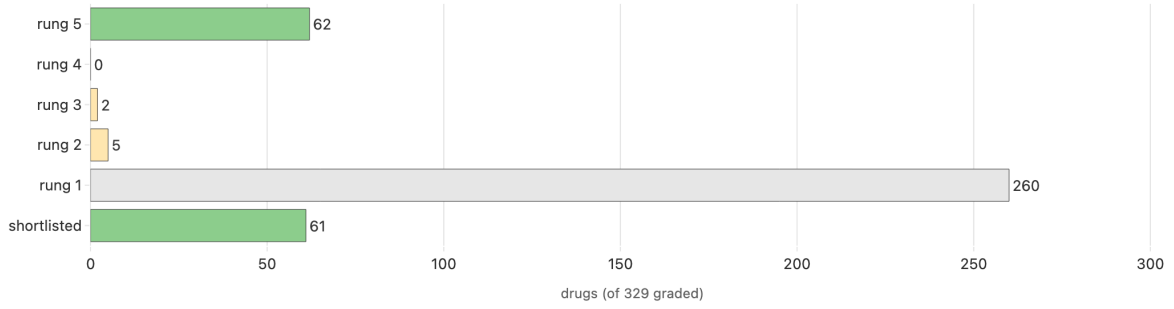
7.1 The eleven drugs with a cited mechanism result, ordered as this report argues them

#	drug (approved for)	direction	rung	the cited result	inherited?	given to children?
1	**lovastatin** (cholesterol)	D4 tumour	5	lovastatin itself slowed rhabdomyosarcoma cells; the tumour cannot take up cholesterol instead of making it (Gizaw 2025, PMID 41249736); a completed phase-1 study gave a statin with chemotherapy to children with relapsed solid tumours including rhabdomyosarcoma (NCT02390843)	no	label 8.4; studies admitting children
2	**simvastatin** (cholesterol)	D4	5	same class result; it is the statin in NCT02390843	yes (lovastatin)	studies admitting children
3	**atorvastatin** (cholesterol)	D4	2	same class result; carries a paediatric indication on its label	yes (lovastatin)	label 8.4; studies
4	**nicotinamide** (vitamin B3)	D3 more BubR1	3	the NAD^+ precursor NMN raised BubR1 in mice; more SIRT2 extended the life of BubR1-deficient mice (North 2014, PMID 24825348)	yes (NMN)	studies admitting children
5	**niacin** (cholesterol; vitamin B3)	D3	3	same result; the prescription-strength NAD^+ precursor	yes (NMN)	studies admitting children
6	**valproic acid** (epilepsy)	D3	2	HDAC2/3 strip the acetyl mark that protects BubR1 from degradation (Park 2017, PMID 28985013; Choi 2009, PMID 19407811) — an HDAC inhibitor would be expected to slow that loss; inferred, not shown	yes (class)	label 8.4; studies
7	**vorinostat** (lymphoma)	D3	2	same class mechanism	yes (class)	label 8.4; studies
8	**chloroquine** (malaria)	D2 stress — literature only	2	chloroquine itself preferentially killed cells with an extra chromosome (Tang & Amon 2011, PMID 21315436)	**no**	studies admitting children
9	**bortezomib** (myeloma)	D2	2	aneuploid human cells depend on faster protein degradation and are more sensitive to proteasome inhibitors (Ippolito 2024, PMID 39247952; Chui 2019, PMID 31530568)	yes (class)	label 8.4; studies
10	**metformin** (diabetes)	D2	2	the AMPK activator AICAR was aneuploidy-selective in the same screen	yes (AICAR)	label 8.4; studies
11	**hydroxychloroquine** (lupus)	D2	2	chloroquine's result, inherited by its clinical analogue	yes (chloroquine)	label 8.4; studies

Why this order and not the ladder's: the ladder ranks by rung, and by rung the statins lead. The report then places D3 ahead of D2 because **D3 acts on the constitutional genotype the child has whatever the burden**, while D2's whole rationale is a population of aneuploid cells that step 8 could not find above 2–3 %

in the sample sequenced. Within D2, chloroquine keeps its molecule-level result; the direction as a whole is a tumour-context and chemoprevention hypothesis argued from published biology, and the report says so wherever it is quoted.

Reading the table honestly: **no** drug has a molecule-level animal result on the checkpoint itself. The deepest mechanistic lead rests on one mouse paper, inherited by two vitamin-B3 forms. The best-supported tumour lead has a completed paediatric trial and a molecule-level cell result. The only molecule-level aneuploidy result is chloroquine's, and it now has no measured target population in this sample.



> **Legend** — drugs counted once at their best rung; **green** rung 4–5 and the shortlist, **yellow** rungs 2–3, **grey** rung 1. Most of the 62 at rung 5 are there by the automated test alone: approved, trialled in rhabdomyosarcoma, used in children.

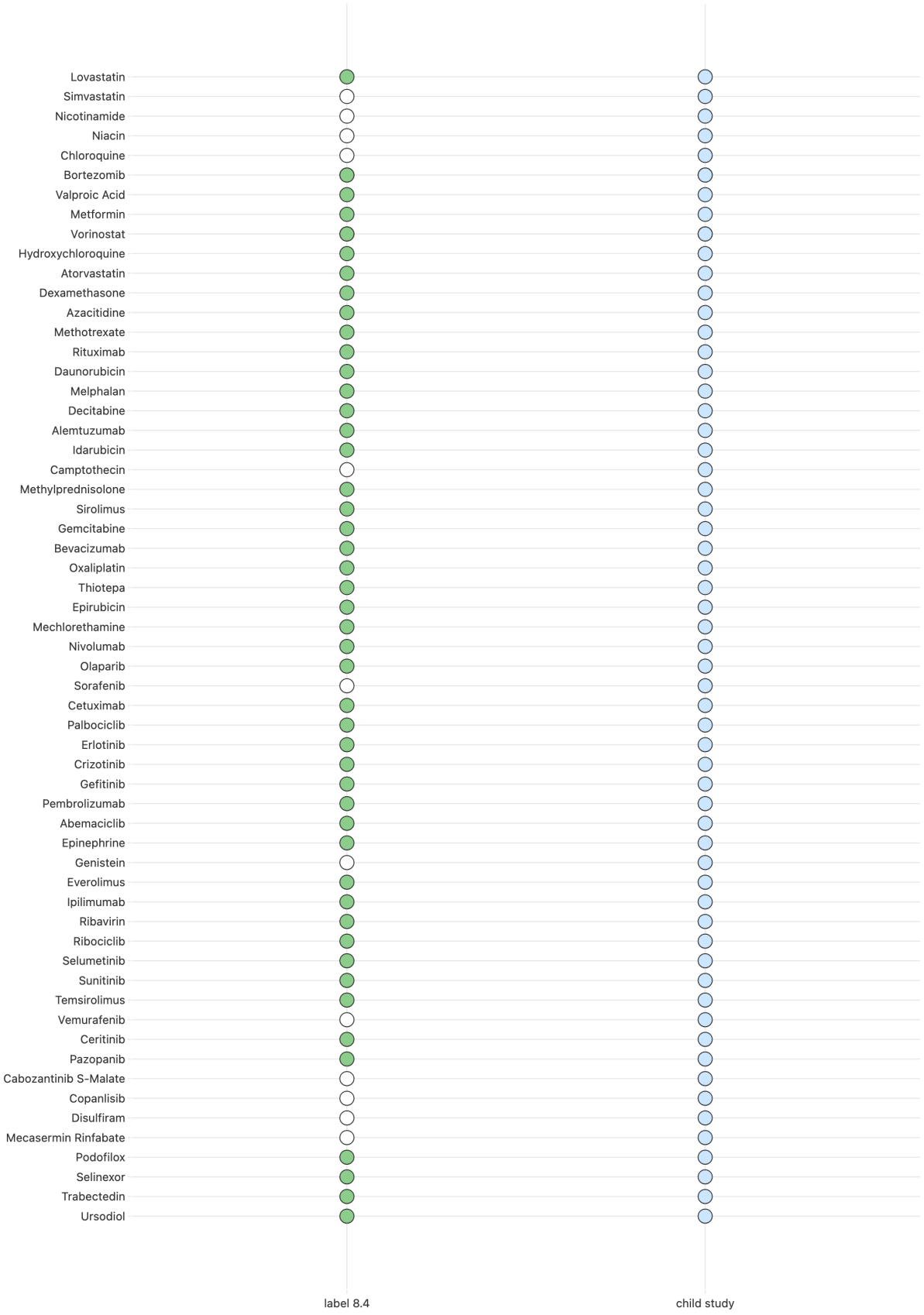
7.2 Fifty drugs already being tried in rhabdomyosarcoma — listed, not claimed

Fifty more approved drugs reach rung 5 automatically (dexamethasone, azacitidine, methotrexate, bevacizumab, the CDK4/6 and mTOR inhibitors, the checkpoint antibodies…). They are in `shortlist.tsv` with the verdict *context trial only – not a mechanism finding*, because the decoy control showed the same list comes out for a gene unrelated to the checkpoint. They are the tumour's trial landscape; a reader deserves to see them; they are not this work's finding.

7.3 The do-not table

verdict	drugs	why
caution — spindle poisons (13)	vincristine, vinorelbine, vinblastine, vinflunine, paclitaxel, docetaxel, cabazitaxel, eribulin, ixabepilone, epothilone D, colchicine, brentuximab vedotin, polatuzumab vedotin	these drugs *rely on the checkpoint to hold the cell*; MVA cells have a weakened checkpoint and slip through (Suijkerbuijk 2010, PMID 20516114; Sinha 2019, PMID 30601084). Vincristine and vinorelbine are standard rhabdomyosarcoma chemotherapy: **a hypothesis for the treating team to weigh, never advice to withhold treatment** — a regimen with survival data behind it outweighs a theoretical mechanism concern
do not — further checkpoint weakening (0 approved)	Mps1/TTK and Aurora inhibitors	aneuploid cancer cells are selectively killed by weakening the checkpoint further (Cohen-Sharir 2021, PMID 33505028) — the same logic makes a child whose checkpoint is constitutionally weak the worst candidate. None is approved; every one in the candidate table is phase 3 or earlier
standard of care (10)	vincristine, dactinomycin, cyclophosphamide, ifosfamide, doxorubicin, etoposide, irinotecan, temozolomide, carboplatin, cisplatin, topotecan	already the treatment for the tumour; giving them is not a repurposing. Named so a reader sees they were considered, not missed

7.4 Has each candidate been given to children?



> **Legend** — **green** a US label with a *Pediatric Use* section (8.4); **blue** at least one registered study admitting children, in any indication; empty = neither. Every argued drug has at least one pointer. **A pointer is something to read, never a licence.**

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8. What this could change — potential impact

> **Rubric:** potential impact (25 %) — for this child, for others living with MVA, and for how MVA is diagnosed. Everything here is conditional on independent validation; nothing in this section has been tested in a patient.

8.1 For this child: one question for the treating team, and one experiment worth running

The do-not table (§7.3) is the part of this work with the shortest path to a clinical conversation, because it needs no new drug — only a question about drugs already in the plan.

The question	Thirteen approved drugs on this child's map are *spindle poisons* (they jam the machinery that pulls chromosomes apart, then **rely on the checkpoint to hold the cell there** until it dies). This child's checkpoint is the part that is broken. **Two of the thirteen — vincristine and vinorelbine — are standard rhabdomyosarcoma chemotherapy.**
Why it is askable now	the genotype is confirmed by the challenge's answer key, not inferred; the mechanism (less BubR1 → a leaky brake → cells slip through mitosis instead of arresting) is published for MVA patient cells (Suijkerbuijk 2010, PMID 20516114; Sinha 2019, PMID 30601084)
What this report says	*weigh it* — a hypothesis about how this class behaves in this genotype, raised early enough to be discussed
What this report does not say	**do not give it.** A regimen with survival data behind it outweighs a theoretical mechanism concern. We have no clinical outcome data, no dose-response in this genotype, and no patient-level evidence of any kind
What would settle it	a checkpoint-function readout (how long cells carrying these allele classes hold mitosis under a spindle challenge, and whether they die or slip) — a bench question, not a chart question

The one experiment we would run next. The deepest mechanistic lead, D3 (raise the amount of working BubR1), rests on rung 3 **by proxy**: a NAD^+ precursor, NMN, raised BubR1 in mice bred to be BubR1-deficient (North 2014, PMID 24825348). That is an animal result, on a related compound, in a different genotype. One experiment would lift it to molecule level in this genotype:

- **Material** — cells carrying this child's allele classes: patient-derived fibroblasts, or an engineered line with one stop allele and one last-exon missense.
- **Treatment** — nicotinamide and niacin (the two approved vitamin-B3 forms this report argues), across a dose range, against vehicle.
- **Read-outs** — (1) BubR1 protein abundance; (2) checkpoint function: how long a cell holds mitosis under a spindle challenge; (3) chromosome missegregation rate.
- **Prediction, written first** — BubR1 abundance rises with dose, and arrest duration lengthens with it.
- **What would refute it** — no rise in BubR1 at any tolerated dose. That result would say the rung-3 animal evidence does not transfer to this genotype, and D3 should be dropped rather than softened.
- **Status — proposal. Not run. No result is claimed for it.** It needs a wet lab and cells this project never had; it is written here in the same pre-registered shape as every claim in §13 so that it can be filed before it is run rather than after.

8.2 For others living with MVA: the conclusions are about the mechanism, not the gene

Three genes are established causes of MVA: `BUB1B` (MVA1) and `TRIP13` (MVA3), both parts of the same mitotic-checkpoint machinery, and `CEP57` (MVA2), a centrosome part (dictionary). Two results in this report are therefore properties of **a weakened mitotic checkpoint**, not of this child's gene:

result	why it is a mechanism property	what it means for another family
no approved drug reinforces the checkpoint (D1 is empty; every checkpoint drug on the market or in trials *weakens* it)	the direction is defined by the pathway, and the pathway is what MVA1 and MVA3 share	the same gap exists for them. It is also a concrete ask for drug developers: the direction this disease needs has **zero** approved entries
the spindle-poison caution	it rests on "the checkpoint holds the cell", which is the mechanism, not the gene	it applies to any constitutionally checkpoint-deficient child facing this drug class, with the same "weigh it, never withhold" framing

This is not a claim we are asking the panel to take on trust — it was **executed**. The pipeline was re-run on a transfer card for `TRIP13` / MVA3 (a different gene, a different disease, a different tumour), and direction D1 returned **30 drugs, every one of them at rung 1** — mechanism only, named and never submitted. The same empty answer, reached again, with no code changed (§10). The honest other half: on that case the spindle-poison caution produced **no** rows at all, for a reason §10 states.

8.3 For diagnosis: a sequencing-only bound on mosaic aneuploidy, with its limit stated

The classical test for MVA is a **karyotype** (counting chromosomes down a microscope in cells first grown in culture). It needs cells that will grow, and it reports a fraction of the cells that grew. Step 8 asks the same question of DNA sequence alone, twice over, and — this is the part that matters — reports **what it could have detected**:

	sequencing-only bound (step 8, two instruments)
what it can do	put a number on how much mosaic whole-chromosome or whole-arm aneuploidy could be hiding in the sequenced sample — no autosome above **2 % of cells** (chr1–6) or **3 %** (chr7–22) at 80 % power by allele balance, and an honest floor of **1.6 % of cells** by k-mer dosage; read **X and Y**, which allele balance cannot in a male, recovering one copy of each (0.4873, 0.5083 against 0.5) as a positive control on a known answer; and do all of it from data already sequenced, with no extra sample and no culture
what it cannot replace	a karyotype, for diagnosis. It is blind to events smaller than a chromosome arm, and the dosage route is blind by construction to a copy-neutral swap (the right number of chromosomes from the wrong parent), which moves no sequence for it to see
what it cannot say	anything about a tissue that was not sequenced. **The delivery never named the tissue of this sample.** MVA is *variegated* — different cells carry different errors — so a bound in one tissue is not a bound in another
what it is, precisely	a **bound**, not a measurement: "nothing above the floor", never "zero"

The transferable point is smaller than a new diagnostic and more useful than it sounds: **a negative result is only a result when it carries its detection limit.** This project's own step 0 produced a flat karyogram with no detection limit; step 8 corrected it on exactly that point (§4). Any pipeline that reports "no aneuploidy found" without a floor is reporting nothing.

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9. What is new here, against standard practice

> **Rubric:** innovation (25 %) — what this pipeline does that a repurposing write-up usually does not, stated as a comparison and hedged to "to our knowledge".

what a repurposing report usually does	what this one does
queries typed by hand into each database, per drug or per gene, and re-typed for the next case	**one structured mechanism card** (`ressources/track2/mechanism.json`: gene, allele classes, pathway, measured facts, directions, seed genes, the ladder) drives **every** query. No query string is typed by hand. The card is versioned and append-only — v3 → v4 → v5, each revision a `history` entry with the entries behind it
produces a ranked list of candidate drugs	produces an **evidence ladder written before the search**, with the **automated** test (approval, label section, registered trials) and the **person-read** test (one PubMed id per row) printed side by side and never merged into one score. Rung 1 is *named, never submitted*
has no negative control — nothing says what the same pipeline returns for the wrong gene	runs a **decoy gene** through the same pipeline and measures the overlap against a threshold fixed beforehand (Jaccard 0.10). D4 failed it at 0.25, so D4 is argued only from curated mechanism results — a direction demoted by its own control (§6.3)
covers one case, and generality is asserted	covers a **transfer case**: a card that swaps the gene, the disease, the tumour and the clinical terms, executed unchanged, with both cases' counts in one committed table (§10)
lists candidates	ships a **do-not table as a first-class output**: caution, do-not, and standard-of-care verdicts, with the reason and the citation per row (§7.3)
asserts confidence after the fact	files each claim in an **append-only register before its test**, with its prediction *and* its refutation condition; confidence is a printed heuristic shown with its arithmetic, never a probability; corrections stay visible — two verdicts committed too early are in the record beside their corrections (§13)

Prior art, named accurately. Knowledge-graph repurposing rankers exist and are strong at the step this pipeline deliberately does not take. **PrimeKG** (a disease-centric biomedical knowledge graph, PMID 36732524) and **TxGNN** (a graph neural network that scores drugs against a disease node, including diseases with no approved drug, PMID 39322717) both take a **disease node** as input and return a ranked drug list. This repository keeps notes on both (`docs/iteration4/cards/primekg.json`, `docs/iteration4/cards/txgnn.json`, both MIT-licensed) and **neither was run for this report.** The difference is the input and the shape of the output: they rank from a disease; this pipeline argues from **one patient's measured mechanism**, and returns a graded argument with a negative control and a do-not list rather than a rank.

How strongly we put it: each ingredient above is standard on its own — public APIs, a grading rubric, a pre-registration habit, a decoy control. What we claim is the **assembly**, and only *to our knowledge*: we have not seen a drug-repurposing write-up that ships a decoy control, a whole transfer case and an append-only claim register together. That is our reading of what we have read, not a literature survey, and we would rather be corrected than overstate it.

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10. Scalability: the same pipeline, a second case

> **Rubric:** scalability (15 %) — executed evidence that this reaches another person, another disease and another

> dataset, together with the pre-registered claim it **refuted**.

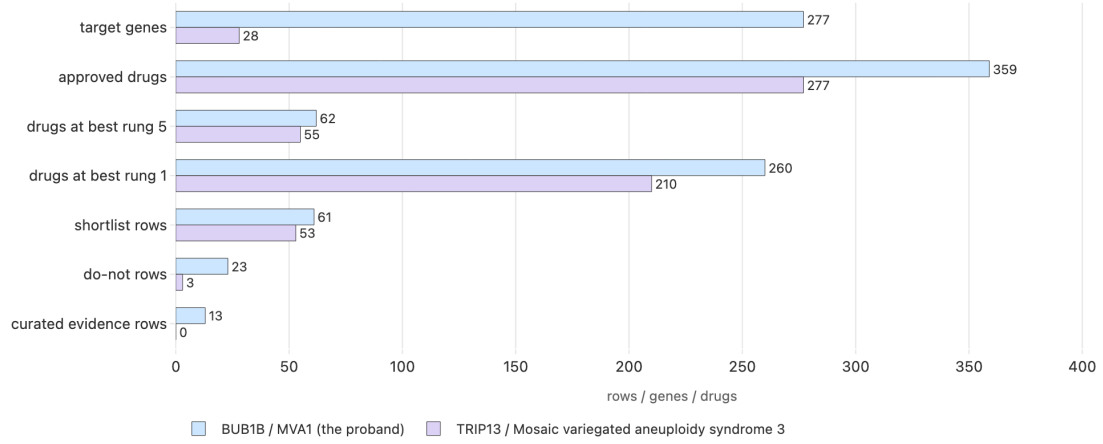
10.1 What was swapped

The decoy control (§6.3) swaps the *gene* and keeps the child: it measures **specificity**. The transfer run swaps the **whole case** and measures **portability** — does the same code, with no edit and no hand-written query, produce the same tables for a case it was never written for?

	the real case	the transfer case
gene	`BUB1B`	`TRIP13`
disease	MVA1 (OMIM 257300, MONDO:0009759)	MVA3 (OMIM 617598, MONDO:0054736)
clinical terms	the 8 HPO terms delivered with the case	**19 terms read out of** the public HPO annotation file (`phenotype.hpoa`, release 2026-09-02) for OMIM:617598 — *read, never typed*; `NOT`-qualified rows skipped
tumour context	rhabdomyosarcoma	nephroblastoma (Wilms tumour)
standard of care	12 drugs	5 drugs, taken only from resolved abstracts
alleles	2 confirmed allele classes	**none** — the transfer card asserts no variant in any person
curated literature	13 person-read rows	**none** — a transfer card has no `evidence.tsv` by construction
copied from the template card	—	exactly **3 case-agnostic blocks**: the anti-directions, the evidence ladder, the eligibility constraints. Everything case-shaped is built from the spec

Nothing was inherited from the child: `mva.drugs.transfer` reads the OMIM id, never the delivery.

10.2 The two cases side by side



> **Legend** — one group per count, one bar per case, **linear** axis and no normalisation: **blue = the real case** (`BUB1B` / MVA1), **purple = the transfer case** (`TRIP13` / MVA3). The point is that both cases produced a table of the same *shape*, so a ratio would hide which is bigger. **`curated evidence rows` is plotted precisely because it is zero** for the transfer case — that zero is the boundary of what this comparison may claim, not a gap in the data. Source: `ressources/track2/transfer_comparison.tsv`, one row per case, with its manifest.

	real case (`BUB1B` / `MVA1`)	transfer case (`TRIP13` / `MVA3`)
target genes	277	28
drug–target candidate rows	3,859	1,329
distinct approved drugs	359	277
drugs at best rung 5 / 3 / 2 / 1	62 / 2 / 5 / 260	55 / 0 / 0 / 210
shortlist rows	61	53 (**every one** `context trial only`, on the tumour direction)
do-not rows	23	3 (all *standard of care*)
curated evidence rows	13	**0**
modules edited to run it	—	**0**

Two results carry over and one does not:

- **D1 is empty again.** All 30 checkpoint-reinforcement rows sit at rung 1 — no submittable drug — and none reaches the shortlist. "No approved drug reinforces the checkpoint" is a property of the mechanism, reproduced on a second gene (§8.2).
- **No trial leaked onto a mechanism direction.** Only the tumour direction D4 reaches rung 2 or higher (55 rows at rung 5); D1 (30), D2 (82) and D3 (39) stay entirely at rung 1 — the narrowing recorded in entry 113 holds on a card it was not written for.
- **The spindle-poison caution did not fire.** The transfer's do-not table holds 3 rows, all standard of care, and **no** row anywhere carries the `A2` flag (real case: 13). §10.5 says why.

10.3 What it cost

	measured
wall clock, a case the pipeline had never seen (cold cache)	**1,693.7 s** end to end, snapshot + enrichment, on the same laptop
live requests, cold	**2,077**; cached replies 0
re-deriving the same tables afterwards (warm cache)	**9.0 s**
money	**€0** — seven public APIs, no cloud, no licensed data source
human time to author the case	one spec file (§10.4). The clinical terms are read from a public file, not written

The asymmetry, stated. The real case's row reads **85 live requests against 1,965 cached** — a *warm* re-run over a cache filled by earlier runs. The transfer's is the *cold* pass that filled its own. **The comparable figure is the transfer's cold pass**, and it is the one quoted whenever this report says what a new case costs.

10.4 What authoring a new case actually takes

One committed spec file. Abbreviated from `ressources/track2/transfer/TRIP13/transfer_spec.json` — the fields a person writes:

```
{
  "gene":   { "symbol": "TRIP13", "uniprot": "Q15645", "hgnc_id": "HGNC:12307",
              "_verified": "UniProt REST + HGNC REST, 2026-09-20" },
  "case":   { "disease": "Mosaic variegated aneuploidy syndrome 3 (MVA3)",
              "omim_phenotype": "617598", "mondo": "MONDO:0054736",
              "tumour_context": "nephroblastoma (Wilms tumour; HP:0002667)",
              "context_standard_of_care": { "drugs": ["VINCRISTINE", "…"],
                                            "source": "PMID 41223336, 42686227" } },
  "repurposing_directions": [
    { "id": "D3", "name": "TRIP13 / p31comet proteostasis axis",
      "rationale": "…a loss-of-function AAA+ ATPase is a protein-remodelling problem…",
      "seed_targets": ["TRIP13", "MAD2L1BP", "VCP", "PSMB5", "HSP90AA1", "HSPA8"],
      "context": false },
    { "id": "D4", "name": "cancer surveillance in the nephroblastoma context",
      "seed_targets": ["WT1", "IGF2", "IGF1R", "CTNNB1", "TP53", "MTOR", "VEGFA"],
      "context": true }
  ],
  "hpo_source": { "_note": "filled in from phenotype.hpoa at build time — read, never typed" }
}
```


What that buys: the disease identifiers drive every disease query; `context: true` marks the one direction the trial-based rungs may grade; `seed_targets` are expanded through Reactome; the phenotype terms are read from a public file. What it does **not** buy: the curated rungs (§10.5). Every identifier in the committed spec was resolved live against UniProt, HGNC, EBI OLS4 and PubMed before the file was committed; anything that would not resolve is recorded as `null` with a note rather than guessed.

10.5 What breaks at scale — including a claim of ours that broke

Our own pre-registered claim came back refuted, and is reported as refuted. H066 said the pipeline is portable *and* the spindle-poison anti-direction would again populate the do-not table. Four of its five limbs held — including every limb that tests portability — and the fourth failed against a refutation condition written before the run. Filed as a conjunction, a false limb refutes it as written: **H066 · refuted · 0.10** (entry 138). The grading heuristic suggested 0.98 — it sums evidence rows and cannot see that a claim is an AND — and that suggestion was rejected on the record.

what breaks	the measured detail	what it means
an anti-direction is only as good as the text it matches	the `A2` flag is a regular expression over each drug's action-type and mechanism-of-action text. Reactome matched **0** pathways for `TRIP13`, so `pathways.tsv` is header-only and all 1,329 candidate rows stay at `target_role = seed`, with only 261 carrying any mechanism text. The real case has **2,276** rows from pathway expansion and 11 of its 13 `A2` drugs arrive on one. Two spindle poisons *are* in the transfer candidate set but reach it from DGIdb with empty mechanism text	the do-not table is a **derived** artefact of how much mechanism text the candidate set happens to carry, not a property of the card. A gene the pathway database cannot expand gets a nearly empty do-not table while the same drugs sit in its candidate set unflagged. **Not fixed here**: widening the flag would change the ladder rule and must be filed as its own pre-registered test, not folded into a refuted claim
the politeness interval on public APIs	a minimum **0.34 s** between live calls (NCBI's 3 requests/second without an API key), single-threaded. 2,077 live requests ≈ **706 s** of enforced waiting — about 40 % of the cold pass	throughput per case is bounded by courtesy, not by compute. Going faster needs API keys and per-source agreements, not a bigger machine
the curated rungs are the human bottleneck	rungs 2 and 3 exist only where a person read a paper: 13 rows for the real case, **0** for the transfer case. Its 53 shortlist rows are therefore *all* `context trial only` — **not one argued drug**	the automated half scales to a new case in 28 minutes; the half that carries the argument does not scale at all yet. A bigger shortlist is not a better one
one drug database left unqueried	**DrugBank** needs a licence for programmatic access; Open Targets, ChEMBL, DGIdb, openFDA, Reactome, ClinicalTrials.gov and PubMed were queried	coverage is what the open sources cover, on both cases equally
what the transfer case did not get	no alleles, no curated literature, no measured facts of its own	**only the automated stages (rungs 1, 4, 5) compare.** `curated_rows` is printed beside every count so the boundary is visible rather than implied
one case, one pathway	`TRIP13` shares the spindle assembly checkpoint with `BUB1B` — chosen so that a transfer that works is a statement about the pipeline, not about how unalike the cases are	portability to a disease *outside* this pathway is untested. It would be a new claim, filed before it is run

The operational rule this buys. A new case's do-not table is only as complete as its pathway coverage. Before the automated safety half of this pipeline is trusted on a new gene, check that the pathway expansion returned something: an empty `pathways.tsv` means the anti-direction flags cannot fire, whatever the card says. That check is a measured boundary of the transfer, found because a refutation condition was written before the run.

What this section does not claim. Not that the transfer case's output is clinically useful — it is a demonstration that the code is case-agnostic, asserts no allele in any person, and is not a drug suggestion for anyone with MVA3. And portability itself is the *finding* of entry 138, resting on the four limbs that held; it is not asserted by H066's verdict, which is a verdict on the conjunction.

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11. What this report cannot tell you

> **Rubric:** scientific rigor (35 %) — the limits of everything above, stated as limits.

1. **Whether any of these drugs helps this child.** Rung 3 is one mouse paper; rung 2 is a dish. Hypotheses for follow-up.

2. **What tissue was sequenced, and therefore what F8 means for the diagnosis.** The delivery does not name it; a karyotype's fraction is measured in cultured cells. F8 says the sequenced sample carries no mosaic population above 2–3 % on any autosome (allele balance) and none above a 1.6 % floor (k-mer dosage, which also reads X and Y and recovers one copy of each). What neither route can see: events **smaller than a chromosome arm**, and a copy-neutral swap that moves no sequence for the dosage route to read.
3. **Whether the weakened BubR1 copy is unstable or inactive.** Outside every catalogued domain; the structure model is confident about the region but silent on function.
4. **Anything about a drug the public sources never listed** beyond what a person put in the curated table — four of the eleven (chloroquine, hydroxychloroquine, niacin, nicotinamide) are there only because a person read the paper; their rows say `curated_only`.
5. **The curated rungs are one reader's**, thirteen abstracts on one day; every row carries its PubMed id so it can be checked.
6. **DrugBank was not queried** (licence); Open Targets, ChEMBL, DGIdb, openFDA, Reactome, ClinicalTrials.gov and PubMed were.

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12. Reproducibility and cost

> **Rubric:** scalability (15 %) and scientific rigor (35 %) — what it costs to re-run, and whether a reader can.

Every step is one command, case-agnostic, with a manifest and a lineage row; no patient data appears in any Track 2 artefact (gene symbols, allele *classes*, drug names, trial ids, PubMed ids only):

```
uv run mva-drugs snapshot                       # card → 5 tables, 7 public APIs, cached repl
uv run python tools/track2_decoys.py ...        # a decoy card and its snapshot (×2)
uv run python tools/track2_enrich.py            # literature, paediatric trials, labels, ladde
uv run python tools/render_track2_figures.py    # the figures in this report
uv run python tools/record_track2_lineage.py    # the lineage edges into the warehouse ledger
uv run python tools/track2_transfer.py          # the whole pipeline again, on a second case (
```

Two laptop runs of about ten minutes each against public APIs, seconds when cached; a case the pipeline has never seen costs **1,693.7 s and 2,077 live requests** end to end, measured (§10). The genotype steps 1–6 and 8 ran on a laptop and one Linux server over a fortnight (their methodology entries state each cost). Code and tables: the public repository named in the form; the dashboard's `/track2` page renders the same tables with the grading visible.

13. Where every claim lives

> **Rubric:** scientific rigor (35 %) — every claim in this report, with its registered prediction, its verdict and its entry.

claim	register id	verdict · confidence	entry
D1 yields no submittable drug and becomes the do-not table	H045	supported · 0.80	113
D2 holds an approved compound with a molecule-level cell result	H046	supported · 0.70	113
D3 has an approved route on the NAD^+/SIRT2 axis with animal evidence, by proxy	H047	supported · 0.60	113
D4 holds a non-chemotherapy approved drug at rung ≥ 2 with paediatric data	H048	supported · 0.80	113
the automated stages are non-specific (D4 overlaps a decoy ≥ 0.10)	H049	supported · 0.85	112
every argued drug has a paediatric pointer	H050	supported · 0.70	113
the ladder is independent of the measured aneuploid fraction (card v4 moved no rung)	H061	supported · 0.85	120
the ladder is independent of the second instrument too (card v5 moved no rung, no shortlist row, no do-not row — three unchanged digests)	**H065**	**supported · 0.90**	**133**
no chromosome carries a mosaic aneuploidy above its own detection limit	H051 (step 8)	supported · 0.80	116
the detection limit is ≤ 10 % of cells for every autosome (measured 2–3 %)	H052 (step 8)	supported · 0.85	116
the two aneuploidy instruments agree on every chromosome within their stated intervals — the k-mer dosage route beside allele balance	**H054** (step 8)	**supported · 0.85**	**121, 123**
the blind fused ranking put the confirmed pair first	H041/H043 (step 6)	see entries	108–109
the pipeline is portable to a different case *and* the spindle-poison caution fires there too	**H066**	**refuted · 0.10** — written as a conjunction; 4 of 5 limbs held (including every portability limb), the caution limb failed because the pathway database returned nothing for the transfer gene. Recorded as refuted rather than reworded	**138**
no approved drug acts directly on BubR1	H011 (iteration 1)	supported · 0.80	078

Confidence is a transparent triage heuristic printed with its arithmetic, never a probability. The register is append-only and shows every revision, including two committed too early and corrected in the open (entries 113 and 120) — and one claim, **H066**, that came back **refuted** and is reported as such rather than softened (§10).